

Mathematical Physics I (Fall 2026): Homework #1 Solution

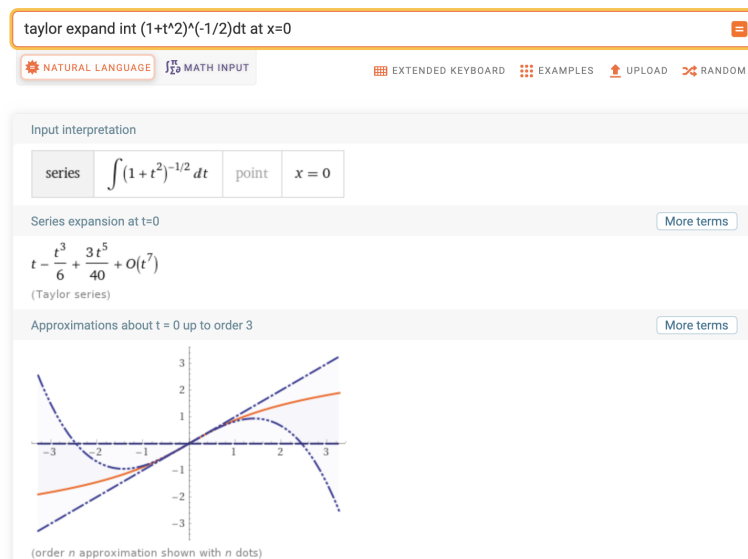
Due Sep. 18, 2026 (Fri, 23:00pm)

[0.5 pt each, total 5 pts]

1. Boas Chapter 1, Problem 13.19

(Note: For Problem 13.19, first prove the equality given, for which you may need to prove and utilize $\sec \theta = \frac{d(\sec \theta + \tan \theta)/d\theta}{\sec \theta + \tan \theta}$.)

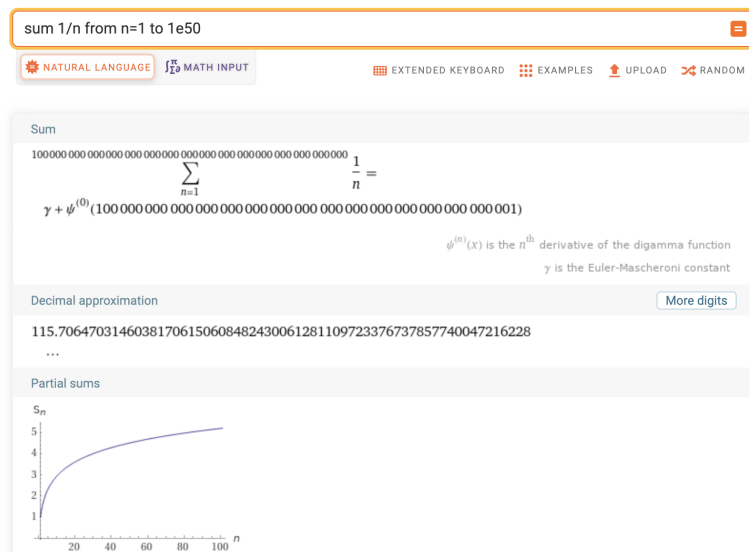
- From Eq.(13.5), you can get the binomial series expansion $(1+t^2)^{-\frac{1}{2}} \simeq 1 + (-\frac{1}{2})t^2 + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}t^4$. Integrating both sides, you find $\int_0^x (1+t^2)^{-\frac{1}{2}} dt \simeq x - \frac{1}{6}x^3 + \frac{3}{40}x^5$. In a manner similar to Example 13.C2 or Problem 13.4, the general terms are found as $\int_0^x (1+t^2)^{-\frac{1}{2}} dt = \sum_{n=0}^{\infty} \frac{(-1)^n (2n-1)!!}{(2n)!! (2n+1)} x^{2n+1}$.



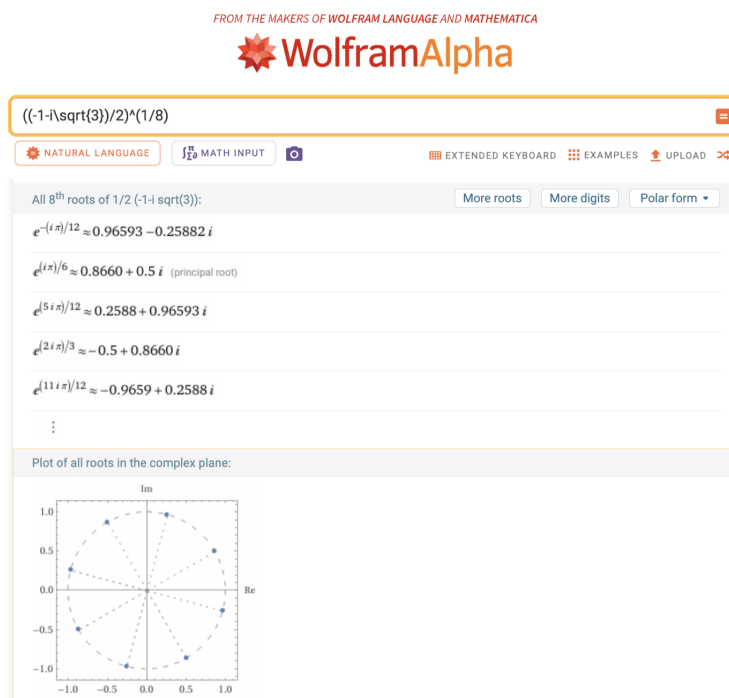
2. Boas Chapter 1, Problem 16.1

(Note: For Problem 16.1, it is helpful to draw a diagram that considers the balance between forces and torques, a so-called *extended free-body diagram*.)

- You may find it interesting to read an article about the stacking problem such as Hall, J., 2005, "Fun with stacking block", American Journal of Physics, 73, 1107.



- The series converges for $-1 \leq x < 1$ according to the ratio test.

$$\bullet \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^{\frac{1}{8}} = \left(e^{\frac{4\pi i}{3} + 2n\pi i}\right)^{\frac{1}{8}} = e^{\left(\frac{\pi}{6} + \frac{n\pi}{4}\right)i} = \left\{ \pm \frac{\sqrt{3}+i}{2}, \pm \frac{1-i\sqrt{3}}{2}, \pm \left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right), \pm \left(\cos\frac{11\pi}{12} + i\sin\frac{11\pi}{12}\right) \right\}.$$


5. Boas Chapter 2, Problem 12.27

- Using Eqs.(11.4) and (12.2), $\sin(4 + 3i) = -\frac{i}{2}(e^{-3+4i} - e^{3-4i}) = -\frac{i}{2}(e^{-3}\cos 4 - e^3\cos 4) + \frac{1}{2}(e^{-3}\sin 4 + e^3\sin 4) = i\cos 4 \sinh 3 + \sin 4 \cosh 3$.

6. Boas Chapter 2, Problem 14.24

(Note: For Problem 14.24, compare your result with a computer solution. Resolve or explain any disagreement in the two results.)

- (a-1) Using the definition of complex powers in Eq.(14.1), $[(-i)^{2+i}]^{2-i} = [e^{(2+i)\ln(-i)}]^{2-i} = [e^{(2+i)i(\frac{3\pi}{2}+2n\pi)}]^{2-i} = [e^{2i(\frac{3\pi}{2}+2n\pi)} \cdot e^{-(\frac{3\pi}{2}+2n\pi)}]^{2-i} = [-e^{-(\frac{3\pi}{2}+2n\pi)}]^{2-i} = e^{(2-i)\ln[-e^{-(\frac{3\pi}{2}+2n\pi)}]} = e^{(2-i)[i(\pi+2m\pi)+\ln[e^{-(\frac{3\pi}{2}+2n\pi)}]]} = e^{2\ln[e^{-(\frac{3\pi}{2}+2n\pi)}]+(\pi+2m\pi)} \cdot e^{i(-\ln[e^{-(\frac{3\pi}{2}+2n\pi)}]+2(\pi+2m\pi))}$.
- (a-2) Meanwhile, $(-i)^{(2+i)(2-i)} = -i$, which can be confirmed with the computer solution below. Note that, from our analytic calculations and computer solutions, your answer in (a-2) is a special case in your answer in (a-1).

 **WolframAlpha**[®] computational intelligence.

{(-i)^(2+i)}^(2-i)
⌵

 NATURAL LANGUAGE
 MATH INPUT
 EXTENDED KEYBOARD
 EXAMPLES
 UPLOAD
 RANDOM

Assuming i is the imaginary unit | Use i as a [variable](#) instead

Input

$((-i)^{2+i})^{2-i}$

i is the imaginary unit

More
Approximate forms
Hide details

General form

$((-i)^{2+i})^{2-i} = e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi n+(1-i/2)\pi}$ for $n \in \mathbb{Z}$

(the choice of n is determined by the branch of the logarithm used for complex exponentiation)

Details

$((-i)^{2+i})^{2-i} = e^{(2-i)\log((-i)^{2+i})}$

$= \exp((2-i)(\operatorname{Log}((-i)^{2+i}) + 2i\pi n))$

$= \exp((2-i)(i \tan^{-1}(\operatorname{Re}((-i)^{2+i}), \operatorname{Im}((-i)^{2+i})) + \frac{1}{2}\log(\operatorname{Im}((-i)^{2+i})^2 + \operatorname{Re}((-i)^{2+i})^2) + 2i\pi n))$

$= e^{(1+2i)\arg((-i)^{2+i})+(2+4i)\pi n+(1-i/2)\pi}$

(Log is the principal logarithm and n is determined by the branch chosen for log)

Values

n	-2	-1	0	1	2
value	$e^{(1+2i)\arg((-i)^{2+i})+(-3-17i/2)\pi}$	$e^{(1+2i)\arg((-i)^{2+i})+(-1-i/2)\pi}$	$e^{(1+2i)\arg((-i)^{2+i})+(1-i/2)\pi}$	$e^{(1+2i)\arg((-i)^{2+i})+(3+7i/2)\pi}$	$e^{(1+2i)\arg((-i)^{2+i})+(5+15i/2)\pi}$

7. Boas Chapter 2, Problem 17.19

- $\arctan z = \omega \rightarrow$ With $u \equiv e^{i\omega}$, you find $z = \tan \omega = \frac{e^{i\omega} - e^{-i\omega}}{i(e^{i\omega} + e^{-i\omega})} = \frac{u - u^{-1}}{i(u + u^{-1})} \rightarrow u^2 = \frac{1+iz}{1-iz}$
 $\rightarrow \omega = \frac{1}{i}\ln u = \frac{1}{i}\ln\left(\frac{1+iz}{1-iz}\right)^{\frac{1}{2}} = \arctan z$.

8. Boas Chapter 3, Problem 2.13

(Note: For Problem 2.13, read the instruction in the textbook carefully; among others, you have been asked to compare your result with a computer solution as well.)

- The rank of the augmented matrix is 2 while there are 3 unknowns.

The screenshot shows the WolframAlpha interface. The input bar contains the text "row reduce {{4,6,-12,7},{5,-2,4,-15},{3,4,-8,4}}". Below the input bar, there are tabs for "NATURAL LANGUAGE" and "MATH INPUT". The "MATH INPUT" tab is selected. Below the tabs, there is a section titled "Input interpretation" which shows the input as a matrix:
$$\begin{pmatrix} 4 & 6 & -12 & 7 \\ 5 & -2 & 4 & -15 \\ 3 & 4 & -8 & 4 \end{pmatrix}$$
. Below this, there is a section titled "Result" which shows the row-reduced matrix:
$$\begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & -2 & \frac{5}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
. There are also buttons for "Decimal form" and "Step-by-step solution".

9. Two coherent waves of the same angular frequency arrive at a detector. In wave optics, the two waves can be described as $y_1(t) = A_1 \cos(\omega t)$ and $y_2(t) = A_2 \cos(\omega t + \phi)$, with amplitudes A_1 and A_2 and a phase difference ϕ . Rather than adding the waves directly using trigonometric identities, one can represent each wave by a complex quantity and take the real part ($\Re[\cdot]$) only at the very end.

(a) Show that the combined total wave can be expressed as

$$y(t) = \Re \left[\left(A_1 + A_2 e^{i\phi} \right) e^{i\omega t} \right],$$

and that the amplitude of the resulting wave becomes $\sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$.

(b) Show that the intensity of the total wave can then be written as $I = I_1 + I_2 + 2\sqrt{I_1I_2} \cos \phi$, where I_1 and I_2 are the intensities of the two individual waves. Find the maximum and minimum possible intensities, $I_{\max}$ and $I_{\min}$, in terms of I_1 and I_2 as the phase difference ϕ varies.

(c) Suppose that the two waves have equal amplitudes, $A_1 = A_2$. Show that your result in (b) reduces to $I = 4I_1 \cos^2 \left(\frac{\phi}{2} \right)$. Explain physically why the intensity can become four times the intensity of the individual wave. Finally, suppose that the two waves have amplitudes in the ratio $A_2 = \frac{1}{2}A_1$. Determine the ratio $\frac{I_{\max}}{I_{\min}}$.

(Note: You may want to review the freshman physics textbooks such as Halliday & Resnick.)

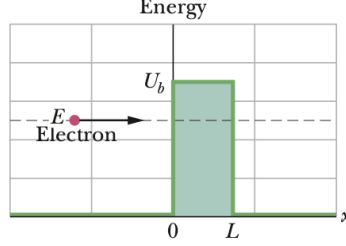
- (b) $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$ and $I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$.
- (c) When with $A_2 = \frac{1}{2}A_1$, $\frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \frac{(1 + \frac{1}{2})^2}{(1 - \frac{1}{2})^2} = 9$.

10. In quantum mechanics, consider the transmission coefficient T of an electron matter wave (of mass m and energy E) incident on a rectangular potential barrier of height U_b (see figure).

Briefly explain why the following expression must be evaluated to determine the probability of the electron being transmitted through the barrier and detected on the other side:

$$T^{-1} = \left| \frac{(k + i\kappa)^2 - (k - i\kappa)^2 e^{-2\kappa L}}{4ik\kappa e^{-\kappa L} e^{-ikL}} \right|^2 = 1 + \left(\frac{k^2 + \kappa^2}{2k\kappa} \right)^2 \sinh^2(\kappa L), \quad (1)$$

where $k \equiv \frac{\sqrt{2mE}}{\hbar}$ and $\kappa \equiv \frac{\sqrt{2m(U_b - E)}}{\hbar}$ are real numbers. Prove the second equality using complex algebra.



[Adapted from Halliday & Resnick, Chapter 38.9]

(Note: You may want to review the freshman physics textbooks such as Halliday & Resnick (Chapters 38.8 and 38.9), or the quantum mechanics textbooks such as Griffiths & Schroeter (Chapter 2.6 and problems therein) or Landau & Lifshitz. To receive full credit, you should include a brief explanation — without a detailed derivation — of how the absolute square of a complex number like Eq.(1) arises in calculating the transmission probability.)

- The wave function $\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & \text{for } x < 0, \\ Ce^{-\kappa x} + De^{\kappa x}, & \text{for } 0 < x < L, \\ Fe^{ikx}, & \text{for } x > L \end{cases}$ describes an electron in the

three regions in the figure. Using the four boundary conditions – continuity of ψ and $\frac{d\psi}{dx}$ at $x = 0$ and L – on five unknowns, one can get the ratios $\frac{D}{C}$, $\frac{B}{A}$, and $\frac{F}{A}$. In particular, one finds the inverse of the transmission coefficient as

$$T^{-1} = \left| \frac{A}{F} \right|^2 = \left| \frac{(k + i\kappa)^2 - (k - i\kappa)^2 e^{-2\kappa L}}{4ik\kappa e^{-\kappa L} e^{-ikL}} \right|^2.$$

- Now with complex algebra,

$$\begin{aligned} T^{-1} &= \left| \frac{(k + i\kappa)^2 e^{\kappa L} - (k - i\kappa)^2 e^{-\kappa L}}{4ik\kappa e^{-ikL}} \right|^2 = \left| \frac{(k^2 - \kappa^2 + 2ik\kappa)e^{\kappa L} - (k^2 - \kappa^2 - 2ik\kappa)e^{-\kappa L}}{4ik\kappa e^{-ikL}} \right|^2 \\ &= \left| \frac{2(k^2 - \kappa^2) \sinh(\kappa L) + 4ik\kappa \cosh(\kappa L)}{4ik\kappa e^{-ikL}} \right|^2 = \frac{(k^2 - \kappa^2)^2 \sinh^2(\kappa L) + 4k^2 \kappa^2 \cosh^2(\kappa L)}{4k^2 \kappa^2} \\ &= \frac{(k^2 + \kappa^2)^2 \sinh^2(\kappa L) + 4k^2 \kappa^2 \{\cosh^2(\kappa L) - \sinh^2(\kappa L)\}}{4k^2 \kappa^2} \\ &= 1 + \left(\frac{k^2 + \kappa^2}{2k\kappa} \right)^2 \sinh^2(\kappa L). \end{aligned}$$